

1:2:2:5 Ref. 李思. Ciccotti Arnold.

Lecture 01 From Newtonian Mechanics to Lagrangian Mechanics.

Particle $\vec{x}: t \mapsto \vec{x}(t) \quad \mathbb{R} \rightarrow \mathbb{R}^3$

Velocity $\vec{v} = \dot{\vec{x}} = \frac{d\vec{x}}{dt}$.

Acceleration $\vec{a} = \dot{\vec{v}}$.

△ Newton Mechanics:

- ① A particle remains at rest or motion at constant velocity, unless it is acted by a force.
- ② $\vec{F} = m\vec{a}$.
- ③ For a group of particles $\vec{x}_i, i=1, \dots, N$.
Exists two kinds of force: External force & Internal force, and $\vec{F}_{ij} = \vec{F}_{ji}$.

For each particle, the motion is decided by

$$m_i \ddot{\vec{x}}_i = \vec{F}_i^e + \sum_j \vec{F}_{ji}$$

and proper initial condition.

☆ General Procedure of Newtonian Mechanics to Solve a Problem

- ① Analyze forces on each particle.
- ② List Newton's equation for each particle.

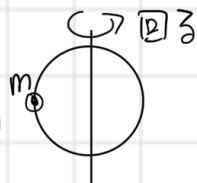
However... (Difficulty).

Ex. Double pendulum 摆

constrained force is unknown in the first place...

Ex. 回环 环

rotating ring



→ "constrained system"

Type of constraints.

- } bilateral $g(\vec{x}, \vec{\dot{x}}, t) = 0$
monolateral $g(\vec{x}, \vec{\dot{x}}, t) \geq 0$
- } time-dependent
time-independent.
- } holonomic 完整约束 (不依赖于 $\vec{\dot{x}}$). $g(\vec{x}, t) = 0$
non-holonomic

In the holonomic case, if we have N particles, k constraints: $g_i(\vec{x}_i, t) = 0, i=1, \dots, k$, the degree of freedom: $3N - k$

We can use $3N - k$ generalized coordinates to describe our system.

$\{q_t\}, t=1, \dots, 3N - k$.

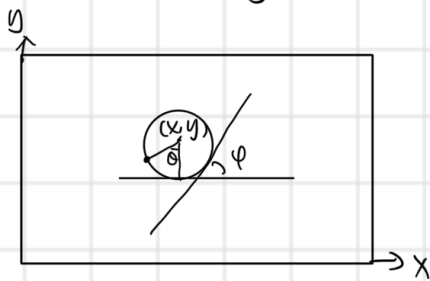
More specifically, we can rewrite $\vec{x}_i, i=1, \dots, 3N$ by using the generalized

coordinates: $\vec{x}_i = \vec{x}_i(q_1, \dots, q_{3N-k}, t)$

$$\frac{d\vec{x}_i}{dt} = \sum_j \frac{\partial \vec{x}_i}{\partial q_j} \dot{q}_j + \frac{\partial \vec{x}_i}{\partial t}$$

For non-holonomic case, constraint equations will contain terms involving $\dot{\vec{x}}_i$. Usually we need more generalized coordinates.

Ex. A non-sliding wheel on a table



$$\begin{cases} r\dot{\theta}\cos\varphi = \dot{x} \\ r\dot{\theta}\sin\varphi = \dot{y} \end{cases}$$

$\Rightarrow$ we still need 4 coordinates to describe the system.

Reason to develop Lagrangian Mechanics.

- ① Newtonian mechanics not suitable for solving problems with complicated constraints.
- ② Lagrangian mechanics is suitable for more general context.

d'Alembert Principle (Virtual Work Principle)

Consider mechanical system in equilibrium, consider at fixed time t , a virtual displacement $\delta\vec{x}_i$. denote force at it as $\vec{F}_i$.

We trivially have $\sum_i \vec{F}_i \cdot \delta\vec{x}_i = 0$.

Separate the force on each particle as

$$\vec{F}_i = \underbrace{\vec{F}_i^e}_{\text{external force}} + \underbrace{\vec{F}_i^c}_{\text{constraint force}}$$

Define constraint as **ideal constraint** if $\sum_i \vec{F}_i^c \cdot \delta\vec{x}_i = 0$.

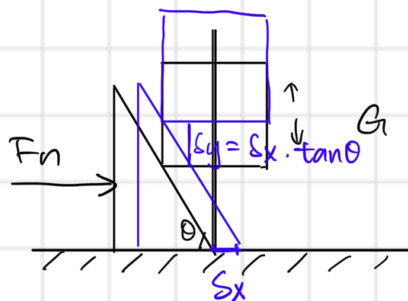
[$\delta\vec{x}_i$ should satisfy certain constraints, otherwise $\vec{F}_i^e$ and $\vec{F}_i^c = 0$].

If All constraints are ideal constraints, then

$$\sum_i \vec{F}_i^e \cdot \delta\vec{x}_i = 0$$

"Virtual work principle"

Ex.

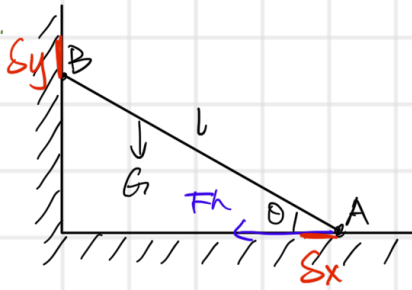


Newton v

Virtual Work:

$$\begin{cases} F_h \cdot S_x - G \cdot S_y = 0 \\ S_y = S_x \cdot \tan\theta \end{cases}$$

Ex.



Virtual Work:

Generalization

$$\sum_i \vec{F}_i^e \cdot \delta \vec{x}_i = 0 \longrightarrow \sum_i (\vec{F}_i^e - m_i \vec{a}_i) \delta \vec{x}_i = 0.$$

つづ'て

Haraki

9.14.